



New Vest, Same Vigilance

A Fourteen-Week Survey of Warehouse Floor Staff Against Their Hi-Vis Vest Tier and Time on the Job

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Background

Hi-vis vest tiers let proximity-sensing machinery tell trainees from time-served staff at a glance. Adoption is near-universal across monitored sites. Whether the tier tracks anything about how safe a worker actually feels on shift has gone untested.

Aims

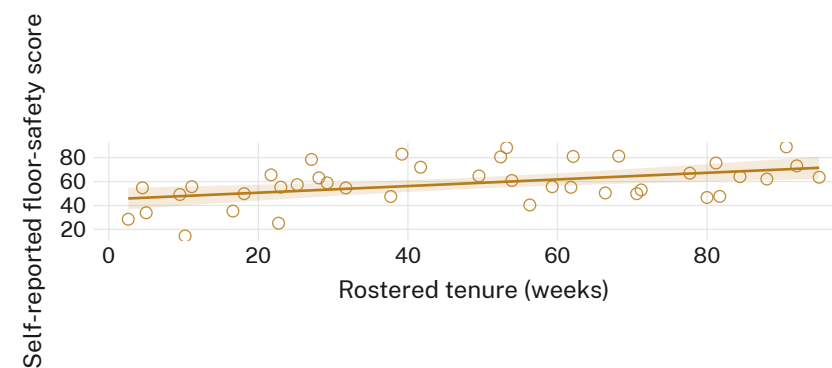
- Test whether vest tier predicts self-reported floor safety
- Compare tier against rostered tenure as a safety predictor
- Establish whether either outperforms a raw near-miss headcount

Methods

- n = 236 floor staff · 19 warehouses · anonymous exit-shift survey · 14 weeks
- Ordinal regression: self-reported safety vest tier + tenure + near-misses
- Tier assignments held constant; no mid-trial re-tiering authorised at any site

Results

Vest tier showed no reliable association with self-reported safety once tenure was controlled for. Tenure alone predicted safety ratings about as well as tier and tenure combined.



Safety score rose with tenure ($r = 0.46$, $n = 236$, 19 warehouses, Apr–Jul 2026); vest tier alone showed none ($r = 0.09$).

Discussion & conclusions

A tier built for machinery reads as the floor’s safety credential — unsupported by what workers feel. Tenure did the tier’s job better than the tier did. Next: does dropping the visible ranking change reported safety, or only who gets asked.

References

1. Christian, M. S., Bradley, J. C., Wallace, J. C., & Burke, M. J. (2009). Workplace safety: A meta-analysis of the roles of person and situation factors. *Journal of Applied Psychology*, 94(5), 1103–1127. doi:10.1037/a0016172
2. Beus, J. M., Bergman, M. E., & Payne, S. C. (2010). The influence of organizational tenure on safety climate strength: A first look. *Accident Analysis & Prevention*, 42(5), 1431–1437. doi:10.1016/j.aap.2009.06.002
3. Seidu, R. K., & Jiang, S. (2025). A review of high-visibility clothing for accident prevention and safety of workers and pedestrians. *The Journal of The Textile Institute*, 116(12), 3285–3298. doi:10.1080/00405000.2025.2491453
4. Giallanza, A., La Scalia, G., Micale, R., & La Fata, C. M. (2024). Occupational health and safety issues in human-robot collaboration: State of the art and open challenges. *Safety Science*, 169, 106313. doi:10.1016/j.ssci.2023.106313
5. Haney, J. M., & Liang, C.-J. (2024). A literature review on safety perception and trust during human-robot interaction with autonomous mobile robots that apply to industrial environments. *IJSE Transactions on Occupational Ergonomics and Human Factors*, 12(1–2), 6–27. doi:10.1080/24725838.2023.2283537

Anonymous exit-shift survey; no individual roster or near-miss record retained beyond the trial window. We thank the floor staff of the nineteen monitored warehouses.



“Same floor. Different vest. Same nerves.”

Self-reported floor safety tracked tenure ($r = 0.46$), not vest tier ($r = 0.09$), across 19 warehouses.